

Test: Observability, Observers, and Observer Based Control

1. Is the following system controllable?

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -3 & 0 \\ 1 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t)\end{aligned}$$

2. Is the following system observable?

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} -3 & 0 \\ 1 & -2 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \\ y(t) &= \begin{bmatrix} 0 & 1 \end{bmatrix} x(t)\end{aligned}$$

3. Consider the one-dimensional system

$$\begin{aligned}\dot{x}(t) &= -x(t) + u(t) \\ y(t) &= x(t).\end{aligned}$$

Design an observer gain L such that the observer has an eigenvalue at -5.

4. Consider the following system

$$\dot{x}(t) = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

with state feedback

$$u(t) = \begin{bmatrix} -2 & -2 \end{bmatrix} x(t).$$

What are the eigenvalues of the closed-loop system?